# Run the Shared sjSDM Tuning Sequence

## Description

Orchestrates isolated unit and tier stores for exhaustive or staged tuning. Staged execution alternates one cumulative unit round with one tier-wide survivor target, then publishes the existing final tier artifact.

## Usage

```
run_sjsdm_tuning_sequence(
  unit_pipeline = NULL,
  tuning_target_names = NULL,
  unit_store_suffixes = NULL,
  prebuild_interpolation = FALSE,
  fresh_run = FALSE,
  tuning_strategy = NULL,
  n_rounds = NULL,
  run_pipeline_function = run_pipeline,
  vec_allowed_profile_roles = base::c("main", "smoke"),
  vec_allowed_profile_statuses = "active",
  has_tuning_evidence_function = has_sjsdm_tuning_evidence,
  target_store = load_active_config_value("target_store"),
  load_metadata_function = load_targets_store_metadata,
  diagnose_endpoints_function = diagnose_sjsdm_cv_preparation_endpoints,
  read_target_function = targets::tar_read_raw,
  invalidate_target_function = targets::tar_invalidate,
  validate_tuning_targets = base::identical(run_pipeline_function, run_pipeline),
  cache_recovery_attempts = 8L,
  validate_calibration_function = validate_sjsdm_cv_production_calibration,
  unit_target_names = NULL
)
```

## Arguments

| | |
|---|---|
| `unit_pipeline` | Character scalar path to the unit pipeline script. |
| `tuning_target_names` | Non-empty character vector of public unit tuning-summary targets. |
| `unit_store_suffixes` | Optional character vector of isolated spatial-unit store suffixes. 'NULL' runs one non-nested unit store, as used by temporal mapped pipelines. |
| `prebuild_interpolation` | Logical forwarded to [run_pipeline()] for unit executions. |
| `fresh_run` | Logical. When true, destroys unit stores on their first round and the tier store before its first aggregation. Later rounds always resume. |
| `tuning_strategy` | Character scalar, '"exhaustive"' or '"staged"'. |
| `n_rounds` | Positive integer configured repeat/tuning-round count. |
| `run_pipeline_function` | Injectable pipeline runner. Defaults to [run_pipeline()]. |
| `vec_allowed_profile_roles` | Profile roles forwarded to [run_pipeline()]. Production callers should retain 'c("main", "smoke")'; dedicated runners authorize other roles. |
| `vec_allowed_profile_statuses` | Profile statuses forwarded to [run_pipeline()]. Production callers should retain '"active"'. |
| `has_tuning_evidence_function` | Injectable fail-closed evidence checker. Defaults to [has_sjsdm_tuning_evidence()]. |
| `target_store` | Root targets-store path. Defaults to the active configuration value. |
| `load_metadata_function` | Injectable metadata loader used for legacy-cache recovery. |
| `diagnose_endpoints_function`, `read_target_function` | Injectable tuning-endpoint diagnostics and target reader. |
| `invalidate_target_function` | Injectable target invalidator used only for incompatible legacy caches. |

| | |
|---|---|
| `validate_tuning_targets` | Logical. Validate tuning artifacts after each production unit round. |
| `cache_recovery_attempts` | Positive integer maximum targeted legacy-cache repairs per unit and round. |
| `validate_calibration_function` | Injectable live calibration guard for main production profiles. |
| `unit_target_names` | Optional named per-unit tuning requests from verified preparation. The exact store-target pairs are persisted for resolution-aware tier pooling. |

## Value

Invisible 'NULL'. Pipeline stores contain the durable results. Unit-level failures are reported and skipped; successful units continue.